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Dynamic Modeling and Stiffness Coefficient Analysis of A 3-DOF Flexible Joint Space Manipulator

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***Abstract*—In order to solve the problems of flexible joint space manipulators in dynamics modeling and simulation, a model with three-dimensional four-link flexible-joint manipulators is taken into account. Considering the nonlinear stiffness coefficient of the torsion-spring, and inertia coupling, the dynamics model of flexible-joint manipulators with revolute joints is established by applying the Lagrange method. The correctness of the model is tested and verified by a PD controller with continuous linear feedback loops. Meanwhile, the influence of different values of the flexible joints stiffness coefficient for system stability is taken into analysis and comparison. The results of simulation analysis show that when the stiffness coefficient K_{stiff} is small, flexible variable of the joints is big, which is very adverse to the stability of the system. On the contrary, when the stiffness coefficient K_{stiff} is bigger, flexible variable of the joints is smaller which is more conducive to control the system.**

Keywords-flexible-joint; space manipulators; dynamics modeling; trajectory tracking control

I. INTRODUCTION

Relative to the rigid body space manipulator, the flexible space manipulator, which can replace the astronauts to complete various complex space missions, has many advantages such as lower energy consumption, higher efficiency and larger operating space etc. It is one of the important development direction of the space robot. Its flexibility mainly comes from the link and the joint. The flexibility of link is from the bending and torsional vibration, the system is severely affected by the bending vibration with lower frequency especially for mechanical arm operating larger load. The flexibility of joint is mainly from the reducer, shaft rod, and transmission gears in the transmission system, which is similar with the flexibility of link in nature. In the case of higher control precision demanded in the end of the mechanical arm, the flexibility of joint have a greater impact than the flexibility of link on the space robot system, which ignored will lead to the instability of the system [1].

Spong [2] simplified the model with flexible joints, and presented the "rotor-torsional spring system" model, which assumed that each joint of the model was a linear torsion spring with constant stiffness coefficient. However, Li's [3] experiment results show that the mechanism of action of joint is more complicated than the linear spring, which was actually showed a highly nonlinear. Sasiadek [4] discusses how, elastic vibration of the joints, coupled with their large rotations and nonlinear dynamics, can also induce structural flexibility of the links. Cao [5] used the neural network method to approximate estimation the unknown manipulator inertia and coriolis-force matrix which was applied in control algorithm. The method, which led to its limitations in the space application, added calculated amount in real-time estimating the unknown parameters. Subudhi [6] studied the method to replace the inverse dynamics with the neural network for slow dynamics model to process the influence of uncertainty factors. Based on Takagi Sugeno model, Park [7] established the mathematical modeling, and designed the fuzzy model reference adaptive control strategy. The control strategy was used to control a single link flexible joint manipulator in tracking a sine wave trajectory based on the joint angle with $\pm 2^\circ$ tracking error.

Ulrich [8] analysed six kinds of control strategy, such as PD controller and so on, and respectively expounds practical characteristics and limitations of the six kinds of controllers for flexible joint manipulator control. In order to research the influence of the nonlinear factors observed in the experiment, the simulation has been carried on in the literature [9]. Simulation results show that the proposed model improves the trajectory tracking control effect to a great extent. The system has a better robustness to the larger uncertainty of actual joint stiffness coefficient, and to the modeling error introduced by nonlinear model. The results are typical whose application prospect is very broad for flexible space system. However, the research model above all is mainly for two link flexible-joint manipulators, in actual space application, it is far from the need of space tasks.

Within this context, the contribution of this paper is to establish dynamics model considering nonlinear stiffness coefficient of torsion spring and inertia coupling based on the space three-dimensional four links flexible-joint manipulator, and to test and verify the correctness of the model by PD controller. Meanwhile, the influence of different values of the flexible joints stiffness coefficient for system stability is taken into analysis and comparison.

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II. Dynamic Modeling of A 3-DOF Flexible-Joint Space Manipulator

A. Kinetic and Potential Energy of the System

Based on the four links flexible-joint manipulator, to define the following generalized coordinates: $\mathbf{q}_m$ is the vector denoting the angular displacements of the motor shaft angles, $\mathbf{q}_f$ is the elastic joint vibration vector, $\mathbf{q}$ is the link angle vector. The following conditions are satisfied.

$$q_i = q_{mi} + q_{fi} \qquad \forall i = 1,2,3 \tag{1}$$

In this text, the model has three-DOF, the Lagrangian is given by

$$L(q_{fi}, q_{mi}, \dot{q}_{fi}, \dot{q}_{mi}) = K(q_{mi}, q_{fi}, \dot{q}_{mi}, \dot{q}_{fi}) - U(q_{fi}) \tag{2}$$

The total kinetic energy of the system is calculated by the following equation:

$$K = \frac{1}{2}\sum_{l=1}^{3}\sum_{k=1}^{l}\sum_{j=1}^{l}\left(trace\left(\frac{\partial T_l}{\partial q_j} I_l \frac{\partial T_l^T}{\partial q_k} \right)\dot{q}_j \dot{q}_k \right) + \frac{1}{2}\sum_{l=1}^{3} I_{ml}\left(\dot{q}_{ml}\right)^2 \tag{3}$$

The first term above is the kinetic energy of links, and the second term above represents the kinetic energy of actuator and transmission system. For revolute joints which are actuated directly by dc motors, part of the kinetic energy of rotors comes from its own rotation, and other part form flexible joint vibration. The kinetic energy of link 0 is zero because it is fixed on the base in this context.

Neglecting the gravitational potential energy in the space application, the potential energy is just only the elastic potential energy U which is also from flexible joint vibration.

$$U = \frac{1}{2}\sum_{j=1}^{3}\sum_{k=1}^{3} k_{jk} q_{fj} q_{fk} \tag{4}$$

Where, k_{jk} is nonlinear stiffness coefficient of the flexible joint.

B. Terms Related to Kinetic Energy

Partial differential $\frac{\partial K}{\partial \dot{q}_{mi}}$ and $\frac{\partial K}{\partial \dot{q}_{fi}}$ about the kinetic energy of the system K are

$$\frac{\partial K}{\partial \dot{q}_{mi}} = \sum_{l=1}^{3}\sum_{j=1}^{l} trace\left(\frac{\partial T_l}{\partial q_j} I_l \frac{\partial T_l^T}{\partial q_i} \right)\dot{q}_j + \sum_{i=1}^{3} I_{mi}\dot{q}_{mi} \tag{5}$$

$$\frac{\partial K}{\partial \dot{q}_{fi}} = \sum_{l=1}^{3}\sum_{j=1}^{l} trace\left(\frac{\partial T_l}{\partial q_j} I_l \frac{\partial T_l^T}{\partial q_i} \right)\dot{q}_j \tag{6}$$

The time t derivation above can be obtained

$$\begin{aligned} \frac{d}{dt}\left(\frac{\partial K}{\partial \dot{q}_{mi}} \right) &= \sum_{l=1}^{3}\sum_{j=1}^{l} trace\left(\frac{\partial T_l}{\partial q_j} I_l \frac{\partial T_l^T}{\partial q_i} \right)\ddot{q}_j \\ &+ \sum_{l=1}^{3}\sum_{j=1}^{l}\sum_{k=1}^{l} trace\left(\frac{\partial^2 T_l}{\partial q_k \partial q_i} I_l \frac{\partial T_l^T}{\partial q_j} \right)\dot{q}_j \dot{q}_k \\ &+ \sum_{l=1}^{3}\sum_{j=1}^{l}\sum_{k=1}^{l} trace\left(\frac{\partial^2 T_l}{\partial q_k \partial q_j} I_l \frac{\partial T_l^T}{\partial q_i} \right)\dot{q}_j \dot{q}_k + \sum_{i=1}^{3} I_{mi}\ddot{q}_{mi} \end{aligned} \tag{7}$$

$$\begin{aligned} \frac{d}{dt}\left(\frac{\partial K}{\partial \dot{q}_{fi}} \right) &= \sum_{l=1}^{3}\sum_{j=1}^{l} trace\left(\frac{\partial T_l}{\partial q_j} I_l \frac{\partial T_l^T}{\partial q_i} \right)\ddot{q}_j \\ &+ \sum_{l=1}^{3}\sum_{j=1}^{l}\sum_{k=1}^{l} trace\left(\frac{\partial^2 T_l}{\partial q_k \partial q_i} I_l \frac{\partial T_l^T}{\partial q_j} \right)\dot{q}_j \dot{q}_k \\ &+ \sum_{l=1}^{3}\sum_{j=1}^{l}\sum_{k=1}^{l} trace\left(\frac{\partial^2 T_l}{\partial q_k \partial q_j} I_l \frac{\partial T_l^T}{\partial q_i} \right)\dot{q}_j \dot{q}_k \end{aligned} \tag{8}$$

In addition, partial differential $\frac{\partial K}{\partial q_{mi}}, \frac{\partial K}{\partial q_{fi}}$ are

$$\frac{\partial K}{\partial q_{mi}} = \sum_{l=1}^{3}\sum_{j=1}^{l}\sum_{k=1}^{l} trace\left(\frac{\partial^2 T_l}{\partial q_{mi} \partial q_k} I_l \frac{\partial T_l^T}{\partial q_j} \right)\dot{q}_j \dot{q}_k \tag{9}$$

$$\frac{\partial K}{\partial q_{fi}} = \sum_{l=1}^{3}\sum_{j=1}^{l}\sum_{k=1}^{l} trace\left(\frac{\partial^2 T_l}{\partial q_{fi} \partial q_k} I_l \frac{\partial T_l^T}{\partial q_j} \right)\dot{q}_j \dot{q}_k \tag{10}$$

C. Term Related to Potential Energy

$\frac{\partial U}{\partial \dot{q}_{mi}}$ and $\frac{d}{dt}\left(\frac{\partial U}{\partial \dot{q}_{mi}} \right)$ are both zero for the total potential energy U not including differential terms. Partial differential $\frac{\partial U}{\partial q_{fi}}$ is

$$\frac{\partial U}{\partial q_{fi}} = \frac{1}{2}\sum_{j=1}^{3} k_{ji} q_{fj} + \frac{1}{2}\sum_{k=1}^{3} k_{ik} q_{fk} = \sum_{j=1}^{3} k_j q_{fj} \tag{11}$$

D. Derivation Process of Dynamics Model

Hence, the Euler–Lagrange dynamics equations of a three-DOF flexible-joint manipulator is

$$\begin{aligned} \mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q},\dot{\mathbf{q}})\dot{\mathbf{q}} + \mathbf{K}_{stiff}\mathbf{q}_f &= \mathbf{0} \\ \mathbf{I}_m\ddot{\mathbf{q}}_m - \mathbf{K}_{stiff}\mathbf{q}_f &= \boldsymbol{\tau} \end{aligned} \tag{12}$$

$\mathbf{M}(\mathbf{q}) \in R^{3\times 3}$ is the symmetric positive-definite link inertia matrix, $\mathbf{C}(\mathbf{q},\dot{\mathbf{q}}) \in R^{3\times 3}$ represents the centrifugal-coriolis matrix, $\mathbf{q} \in R^{3\times 3}$ is the link angle vector, $\boldsymbol{\tau} \in R^{3\times 1}$ represents the control torque vector, $\mathbf{I}_m \in R^{3\times 3}$ denotes the positive-definite motor inertia matrix and where the stiffness of the flexible joints is modeled by $\mathbf{K}_{stiff}$, the diagonal positive-definite stiffness matrix of the joints.

III. Simulation and Analysis

A. Controller

In order to verify accuracy of the model, the PD controller is selected to simulate on the conditions of ignoring gravity and external disturbance, and knowing the joint planning trajectory.

$$\boldsymbol{\tau} = -\mathbf{K}_p\mathbf{e} - \mathbf{K}_d\dot{\mathbf{e}} \tag{13}$$

The tracking error is given by $\mathbf{e} = \mathbf{q}_m - \mathbf{q}_d$, $\dot{\mathbf{e}} = \dot{\mathbf{q}}_m - \dot{\mathbf{q}}_d$. It is easy to prove the system has a unique equilibrium point

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defined by $(\mathbf{e},\dot{\mathbf{e}})=(\mathbf{0},\mathbf{0})$. Moreover, this equilibrium is globally asymptotically stable.

B. Numerical Simulation

In order to prove the effectiveness of the designed controller, this paper used the dynamics model of flexible-joint space manipulator to simulate the trajectory tracking control. Model parameters are shown in TABLE I.

The control law

$$\boldsymbol{\tau}=-\mathbf{K}_p(\mathbf{q}_m-\mathbf{q}_d)-\mathbf{K}_d(\dot{\mathbf{q}}_m-\dot{\mathbf{q}}_d) \tag{14}$$

Where $\mathbf{K}_p=diag(3000)$, $\mathbf{K}_d=diag(1000)$. The initial numerical value of simulation is

$\mathbf{q}_{1..3}=[0.1\ 0.08\ 0.12]^T$;

$\mathbf{q}_{m_{1..3}}=[0.20\ 0.15\ 0.12]^T$;

$\mathbf{v}_{1..3}=[0\ 0\ 0]^T$;

$\mathbf{v}_{m_{1..3}}=[0\ 0\ 0]^T$;

$\mathbf{I}_m=diag[0.10097,0.27122,0.2046]$;

$\mathbf{K}_{stiff}=diag[500]$.

Desired trajectory of the joint-angles is respectively:

$$q_{1d}=\frac{1}{10}\sin(\frac{\pi}{10}t)\text{ ;}$$

$$q_{2d}=\frac{1}{10}\sin(\frac{\pi}{10}t)\text{ ;}$$

$$q_{3d}=\frac{1}{10}\sin(\frac{\pi}{10}t)\text{ .}$$

The unit of joint angle, angular velocity and acceleration are rad , rad/s , rad^2/s , moment of inertia is $kg\cdot m^2$, joint stiffness coefficient is $N\cdot M/rad$.

By numerical simulation, the joint angle curves, flexible variable curves, control torque curves and joint trajectory tracking error curves are shown in Fig.1- Fig.4. Due to the torque limiter of each joint motor, when the motor control torque is greater than $30N\cdot M$, the output of motor torque will be limited. From Fig.1- Fig.4, we can see that the controller, which can realize fast and accurate tracking control, has good control effect for this derived model with smaller joint trajectory tracking error, easily to implement the control torque. The flexible vibration of each joint is decayed step by step and within a short period of time it reached to zero.

TABLE I. **PARAMETERS OF SPACE MANIPULATOR**

Parameter	Link 0	Link 1	Link 2	Link 3
Mass (kg)	1000	0	10	8
$\mathbf{a}_i$ (mm)	649	3.40	499.80	8.80
	15	-1.05	8.70	162.40
	-96	93.25	-94.80	-329.50
$\mathbf{b}_i$ (mm)		-123.40	485.20	-8.80
		1.00	-8.68	-42.40
		29.60	94.83	-312.47
Ixx($kg\cdot m^2$)	310	0.16	0.11	1.73
Iyy($kg\cdot m^2$)	303	0.24	3.73	1.44
Izz($kg\cdot m^2$)	152	0.11	3.65	0.32

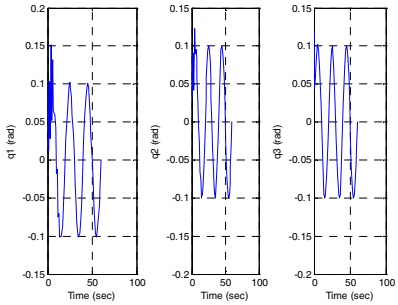


Figure 1. Joint angle curves

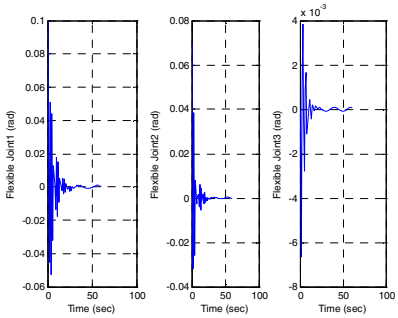


Figure 2. Joint flexible variable curves

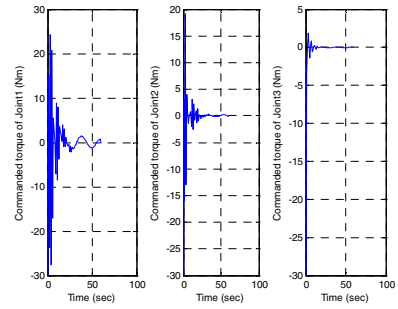


Figure 3. Control torque curves

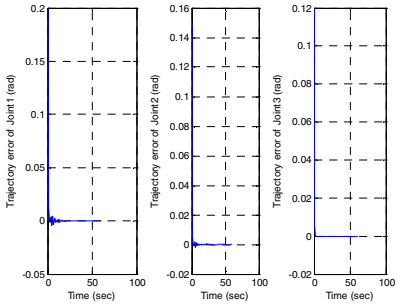


Figure 4. Trajectory tracking error curves

To analyse in-depth the effects of joint flexibility on the system stability, two sets of simulation were carried out respectively by $\mathbf{K}_{stiff}=diag[1000]$ and $\mathbf{K}_{stiff}=diag[100]$. The different values of affects directly the size of the flexible

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joint variable by analysis and comparison, as shown in Fig.5 and Fig.6. When the stiffness coefficient $\mathbf{K}_{stiff}$ is small, flexible variable of the joints is big. The resonance frequency of the system is very low at this time, which reduce greatly the effective bandwidth and is very adverse to the stability of the system. On the contrary, when the stiffness coefficient $\mathbf{K}_{stiff}$ is bigger, flexible variable of the joints is smaller which is more conducive to control the system. Therefore, principle of the control strategy of flexible joint is to solve the problem how to design an appropriate control method to reduce or even to eliminate the influence of the flexible variable, and to enhance the stability of system performance when the flexible variable is too large.

IV. CONCLUSION AND PROSPECT

Within this context, the contribution of this paper is to establish dynamics model considering nonlinear stiffness coefficient of torsion spring and inertia coupling based on the space three-dimensional four links flexible-joint manipulator. Compared with 2-DOF planar space manipulator, the model which has 3-DOF can complete the space mission better. The correctness of the model is tested and verified by a PD controller with continuous linear feedback loops. The results of simulation analysis show that when the stiffness coefficient $\mathbf{K}_{stiff}$ is small, flexible variable of the joints is big, which is very adverse to the stability of the system and a better controller with higher precision needs to be designed compared with the PD controller. On the contrary, when the stiffness coefficient $\mathbf{K}_{stiff}$ is bigger, flexible variable of the joints is smaller, which is more conducive to control the system. The next work mainly include: first and foremost, a more accurate dynamics model considering the external interference, parameter and structure uncertainty need to set up for the further research. After that, in order to solve the problem of PD controller with too large initial output torque, to design a better controller which can realize a faster and more precise position control of the end-effector is very necessary, so that the space operation task could be completed smooth. Finally, how to reduce or even to eliminate the adverse effect of the flexible variable on the stability of the system is still need a hard work.

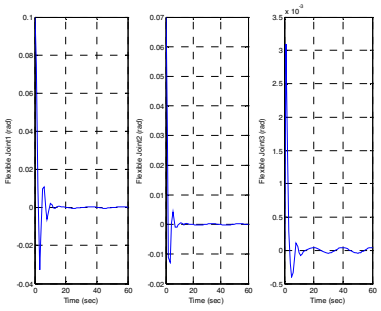


Figure 5. Joint flexible variable ($\mathbf{K}_{stiff} = diag[1000]$)

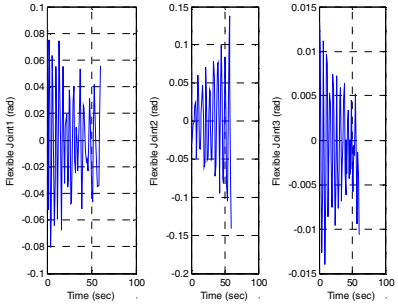


Figure 6. Joint flexible variable ($\mathbf{K}_{stiff} = diag[100]$)

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